SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

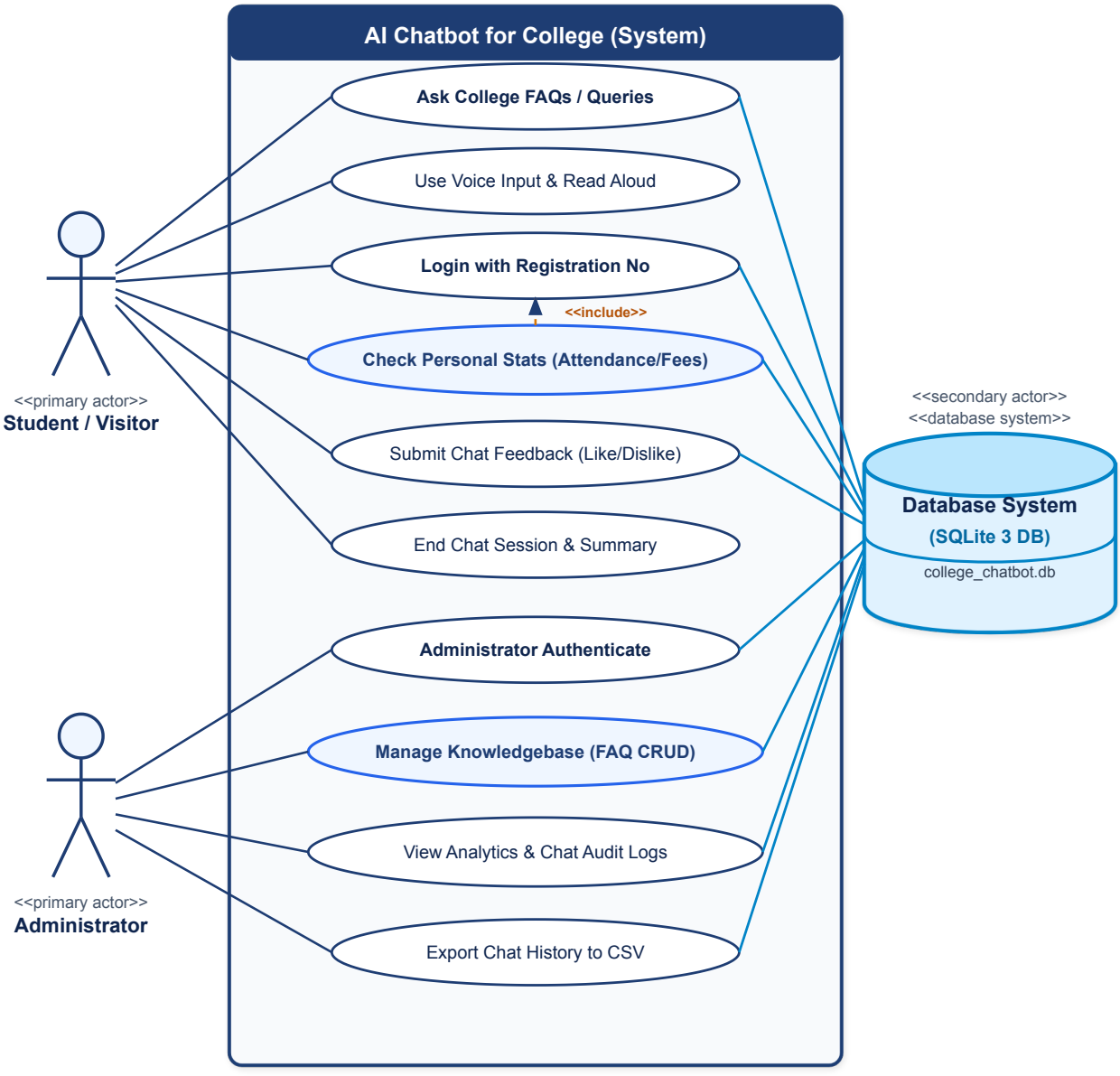


Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and broad secondary database cylinder (SQLite 3 DB) with data-tier associations.

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2. Class Diagram (With Full Relations & Multiplicities)

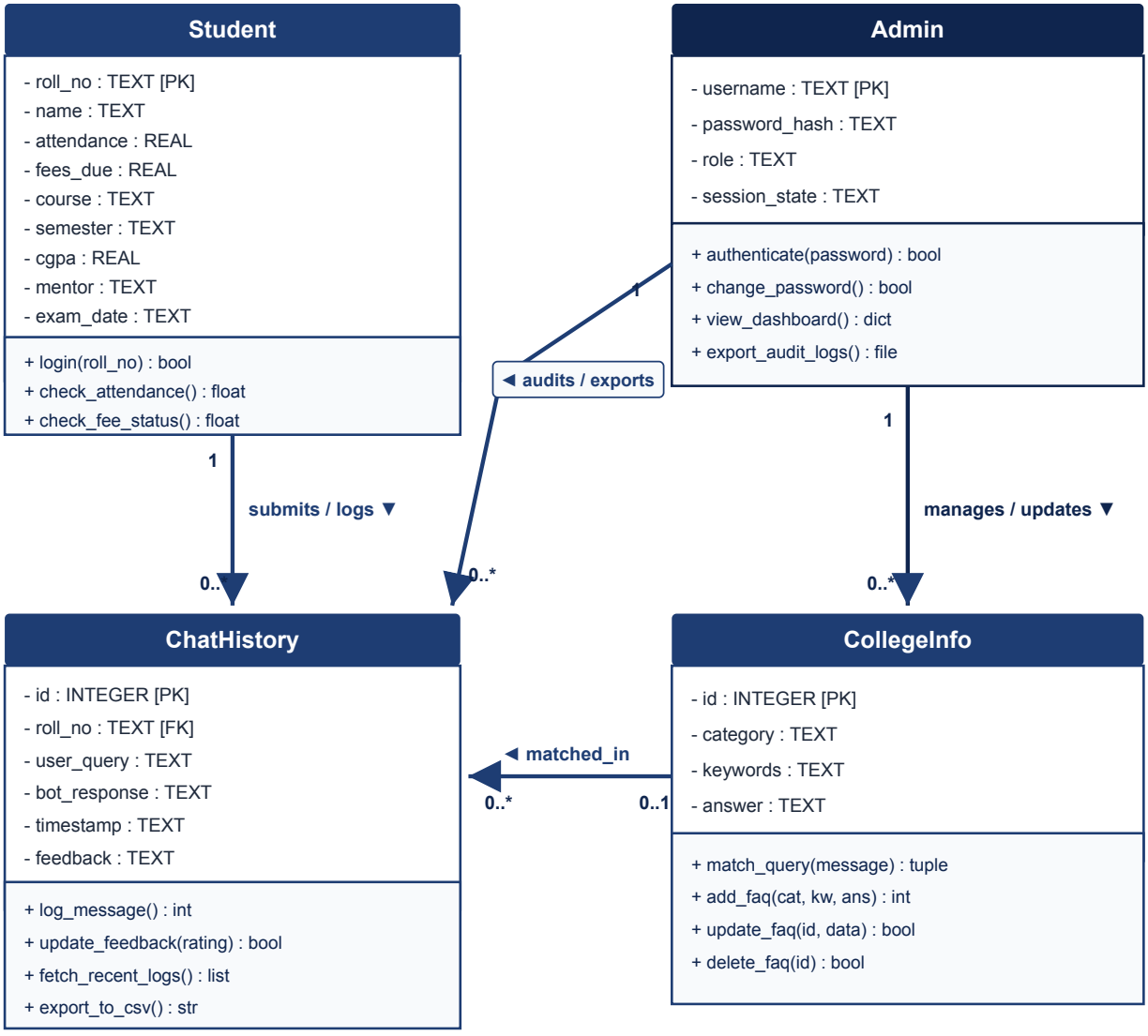
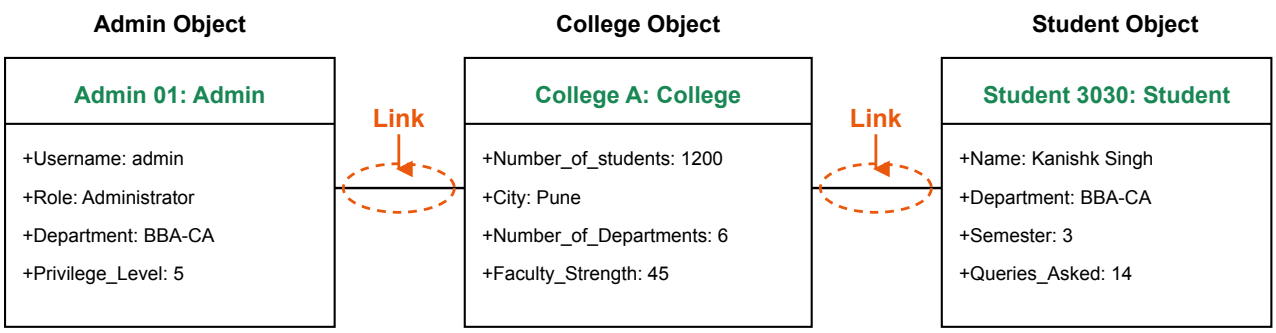


Fig 2: Class Diagram — core entities, attributes, operations, and 100% solid association arrows with multiplicities and role verbs.

3. Object Diagram (Runtime Instances & Slot Values)

An object diagram using a link and 3 objects



An object of class Admin and an object of class Student are linked to an object of class College.

Object Diagrams | Unified Modeling Language (UML) ⚙ GfG

Fig 3: Object Diagram — runtime object instances (Admin Object, College Object, Student Object) connected via formal UML links matching GeeksforGeeks standard notation.

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4. Sequence Diagram — Query, Feedback & Session Termination

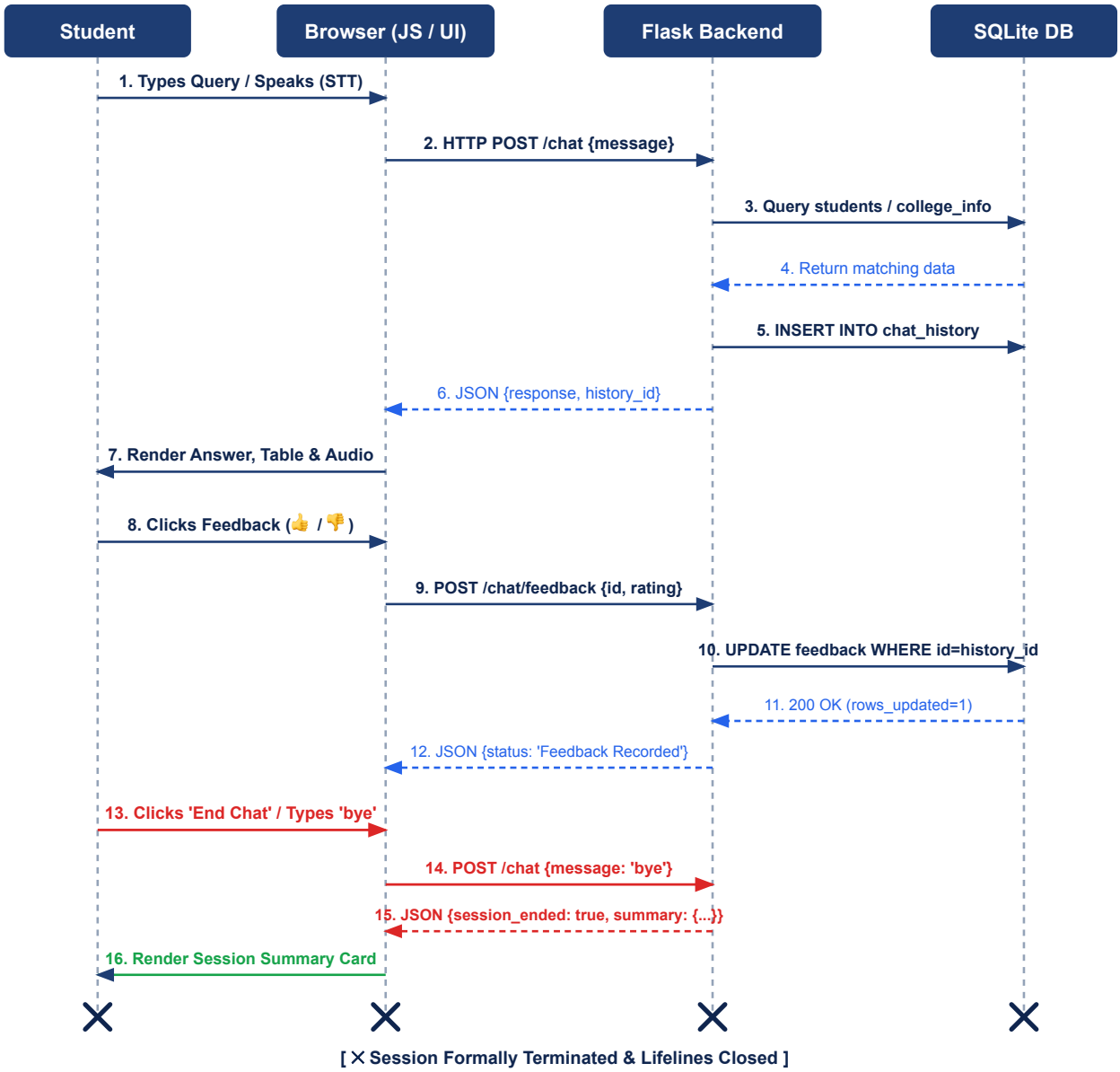


Fig 4: Sequence Diagram — complete query, feedback ACK, explicit End Chat flow, and standard UML lifeline termination (✕).

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5. Activity Diagram — Chatbot Response & Session Flow

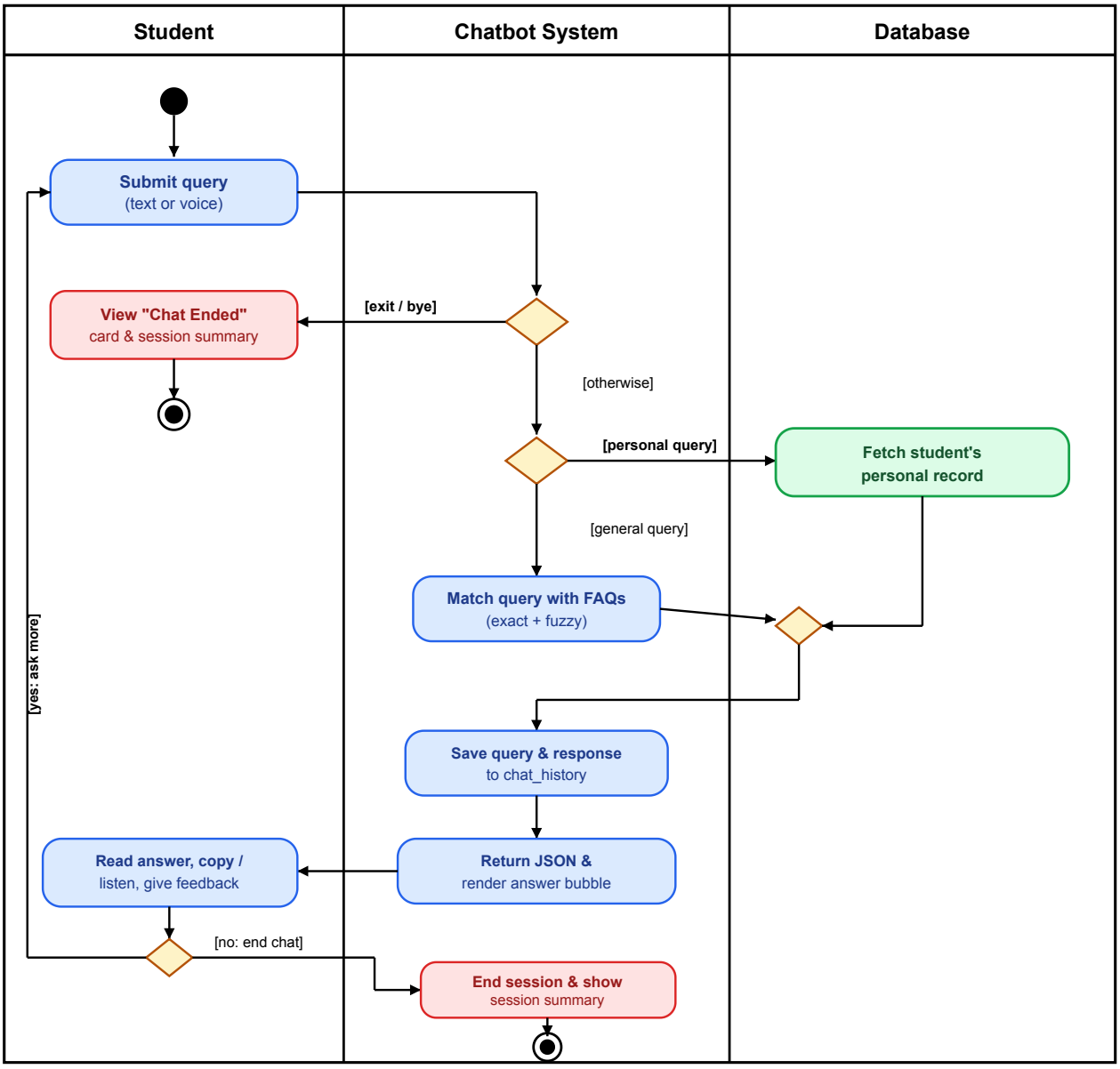
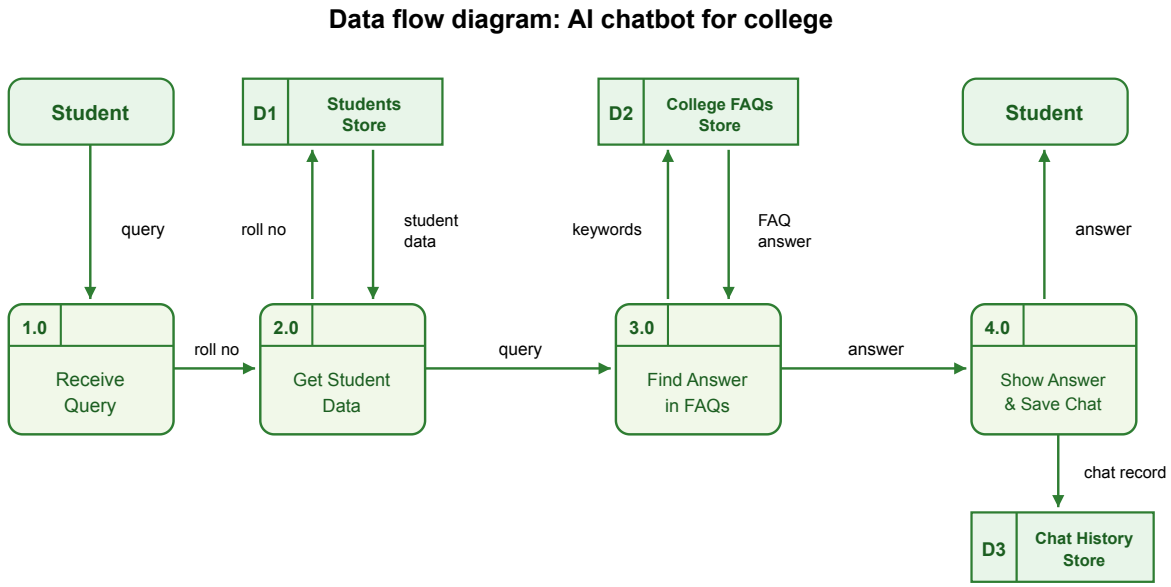


Fig 5: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) explicitly showing database fetching of student personal records, FAQ matching, and controlled session loop with explicit End states.

6. Data Flow Diagram (DFD Level 0 Context & Level 1 Detailed)



DFD symbols and their uses

Symbol	Name	Use	Example in our project
	**External entity**	A person outside the system who gives or gets data	Student
	**Process**	A step that works on data	Find Answer in FAQs
	**Data store**	A place where data is kept	College FAQs Store
	Data flow	An arrow showing data moving from one place to another	query, answer, roll no

Fig 6: Data Flow Diagram (DFD) — First-Level Process Decomposition (1.0 Receive Query to 4.0 Show Answer) with data stores (D1 Students, D2 FAQs, D3 Chat History) and formal DFD symbols reference table.

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7. Entity-Relationship Diagram (ERD — Database Architecture)

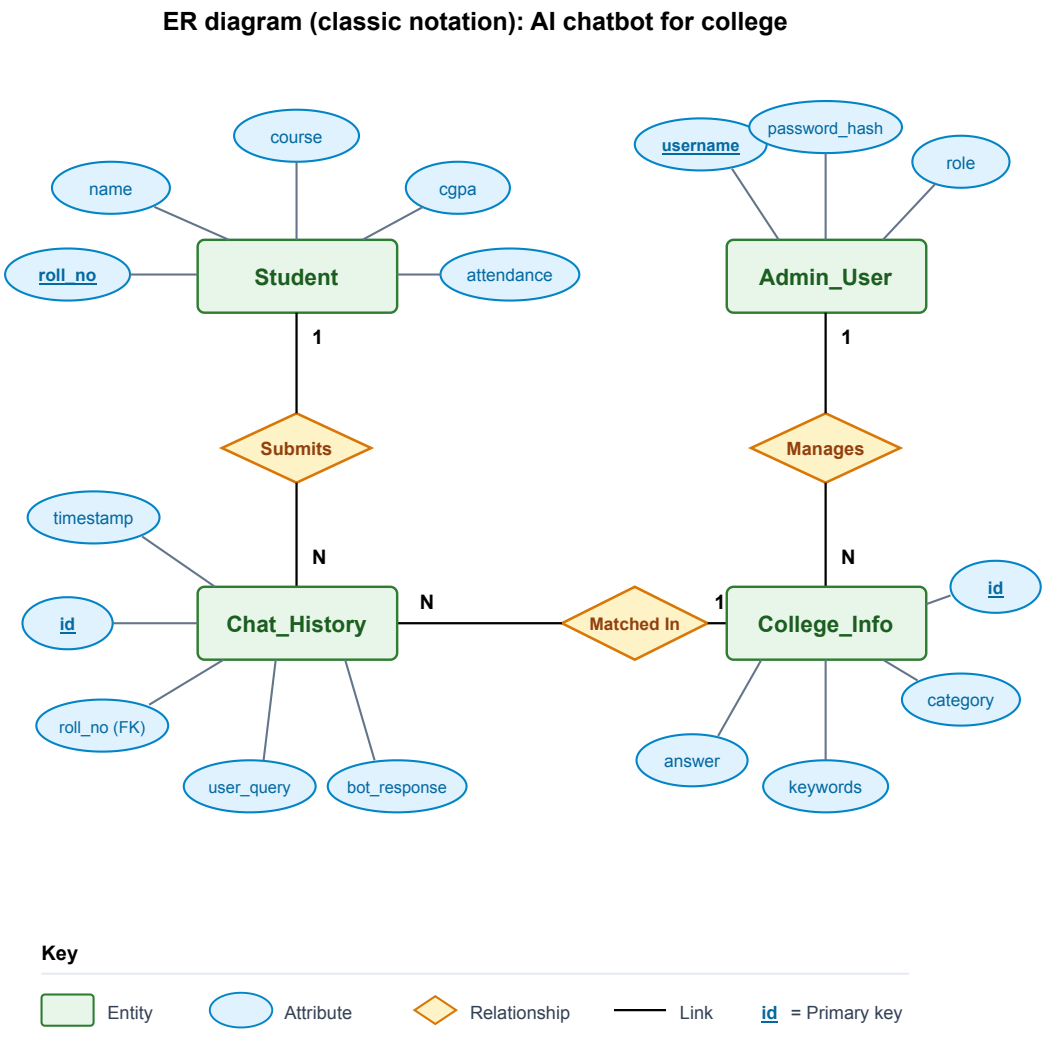


Fig 7: Entity-Relationship Diagram (ERD — Classic Notation) — 4 core entities (Student, Admin_User, Chat_History, College_Info) with attributes, cardinalities (1:N, N:1), and official symbol key.

8. Deployment Diagram (Hosting Topology & Artifact Architecture)

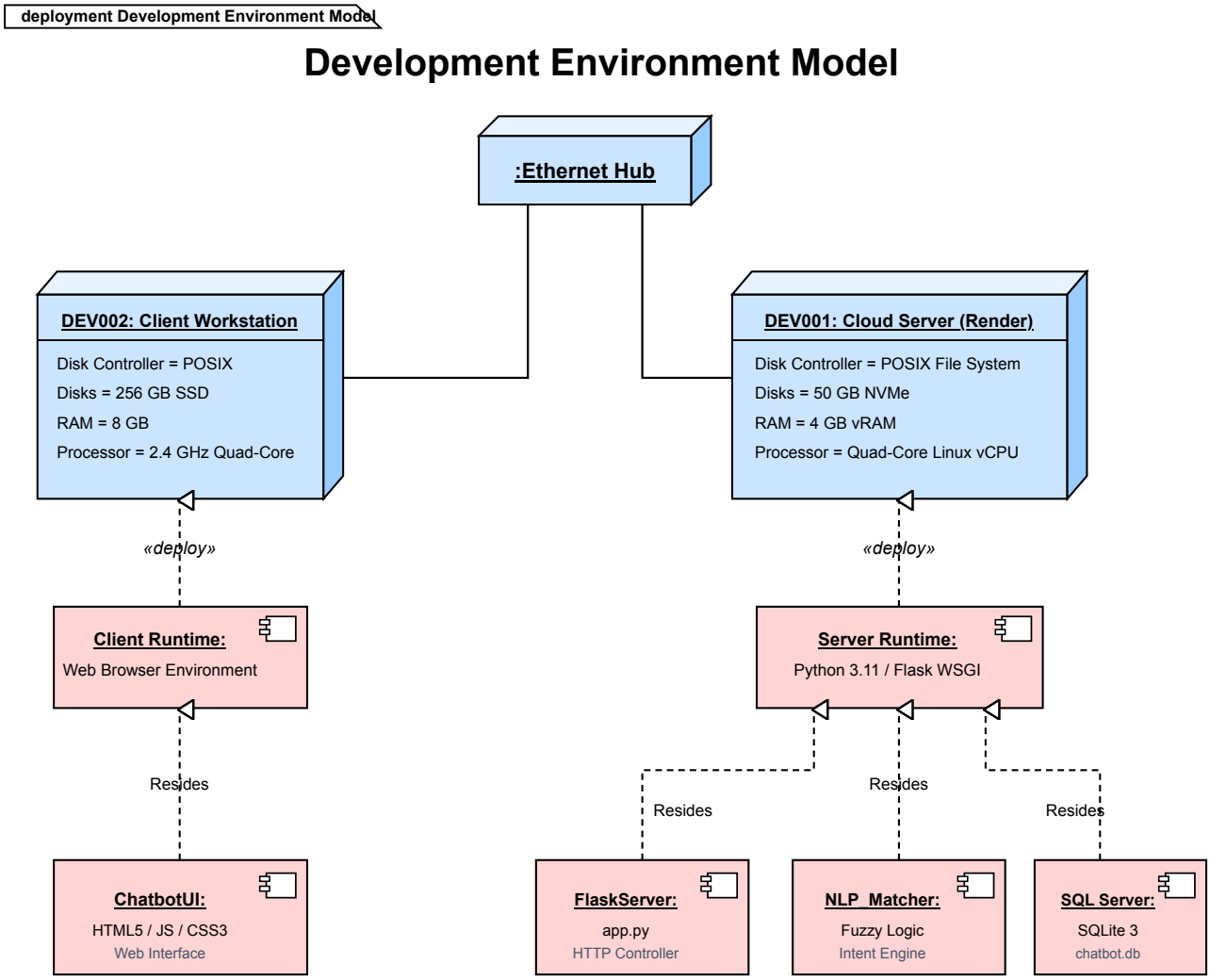


Fig 8: Deployment Diagram — Development Environment Model with :Ethernet Hub, 3D hardware nodes with specifications, «deploy» dependency arrows, and resident software component boxes.